

AP Calc Lesson 2.6 Homework

Answer Key

1. All of the above.

2. $2e^x - \frac{5}{x} + 4^x \cdot \ln 4$

3. $y' = 5e^x + 7$ so $y'|_{(0,5)} = 5e^0 + 7 = 12$, which means that $y = 12x + 5$ is the equation of the tangent line. $0 = 12x + 5 \rightarrow x = -\frac{5}{12}$

4. $a = 2$ and $b = 5$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (5e^x - ax) = 5e^0 - a(0) = 5 \text{ and } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (3x + b \cos x) = 3(0) + b \cos(0) = b = 5$$

$$\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^-} (5e^x - a) = 5e^0 - a = 5 - a \text{ and } \lim_{x \rightarrow 0^+} f'(x) = \lim_{x \rightarrow 0^+} (3 - b \sin x) = 3 - b \sin(0) = 3$$

$$\text{so, } 5 - a = 3 \rightarrow a = 2$$

5. a. $g'(x) = 5 - \frac{6}{x}$ and $f'(3) = -4$ so $5 - \frac{6}{x} = -4 \Rightarrow -\frac{6}{x} = -9 \Rightarrow x = \frac{2}{3}$

b. $h(1) = 2g(1) - f(1) = 2(5(1) - 6 \ln(1)) - 1 = 2(5 - 6(0)) - 1 = 9$

$$h'(x) = 2g'(x) - f'(x) = 2\left(5 - \frac{6}{x}\right) - 3 = 7 - \frac{12}{x} \text{ so } h'(1) = 7 - \frac{12}{1} = -5$$

The equation of the tangent line is given by $y = -5(x - 1) + 9$ or $y = -5x + 14$

6. $y' = \frac{8}{x}$ and $m = \frac{8 \ln x - 0}{x - 0}$ so $\frac{8}{x} = \frac{8 \ln x}{x} \Rightarrow 8x = x(8 \ln x) \Rightarrow 8x - 8x \ln x = 0 \Rightarrow 8x(1 - \ln x) = 0$

$$\text{Since } x \neq 0, (1 - \ln x) = 0 \Rightarrow \ln x = 1 \Rightarrow x = e \text{ so } m = \frac{8}{e}$$

7. $g'(x) = -2 + \frac{4}{x}$ and $\frac{f(5) - f(0)}{5 - 0} = \frac{9 - (-6)}{5 - 0} = 3$, so $-2 + \frac{4}{x} = 3 \Rightarrow \frac{4}{x} = 5 \Rightarrow x = \frac{4}{5}$

8. $f'(x) = \frac{e^x}{2} - 1$ and the slope of the given line is $\frac{1}{2}$, so $\frac{e^x}{2} - 1 = \frac{1}{2} \rightarrow \frac{e^x}{2} = \frac{3}{2} \rightarrow e^x = 3 \rightarrow x = \ln 3$
and $f(\ln 3) = \frac{e^{\ln 3}}{2} - \ln 3 = \frac{3}{2} - \ln 3$, so the point of tangency is $\left(\ln 3, \frac{3}{2} - \ln 3\right)$.

9. a. $f(3+h) > f(3)$. This is because as x increases, $\ln x$ increases.
- b. $f'(3) > f'(3+h)$. This is because $f'(x) = \frac{1}{x}$, which means $f'(3) = \frac{1}{3}$
and $f'(3+h) = \frac{1}{3+h} \cdot \frac{1}{3} > \frac{1}{3+h}$

10. B

11. B